

Accounting Ratio Analysis & Strategic Recommendations

Executive Memo — Caterpillar Inc. (NYSE: CAT) | FY2024

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A. Company and Data Summary

Caterpillar Inc. (NYSE: CAT) is the world's leading manufacturer of construction and mining equipment, operating across 190+ countries. This analysis covers FY2024 (ended December 31, 2024) using CAT's 10-K filing with the SEC. FY2023 balance sheet figures are used for prior-year comparators. Share price is from Yahoo Finance; WACC is an analyst estimate.

Assumption	Value	Basis
Tax rate	21.4%	Statutory rate; not derived from the income statement
WACC	7.9%	Analyst estimate; no formal CAPM build-up
Share price	\$349.58	Fiscal year-end closing price; 492M basic shares
Total capitalization	LT debt+equity	Excludes short-term debt (\$4.4B) and current LTD (\$6.7B)
Quick ratio recv.	Trade only	Cat Financial finance receivables (\$9.6B) excluded

B. Ratio Results Summary

Performance		Profitability	
Market Value Added (MVA)	\$152,499M	Return on Assets (ROA)	14.9%
Market-to-Book (M/B)	8.82x	Return on Capital (ROC)	29.5%
Economic Value Added (EVA)	\$9,518M	Return on Equity (ROE)	59.7%
		ROA — avg. based	14.8%
		ROC — avg. based	28.6%
Efficiency		Leverage & Liquidity	
Asset Turnover	0.74x	LT Debt Ratio	58.4%
Receivables Turnover	6.96x	LT Debt-to-Equity	1.40x
Avg Collection Period	52 days	Total Debt Ratio	77.8%
Inventory Turnover	2.61x	Times Interest Earned	8.28x
Days in Inventory	140 days	Cash Coverage	9.11x
Profit Margin	18.0%	NWC to Assets	15.3%
Operating Profit Margin	20.0%	Current / Quick / Cash	1.42 / 0.50 / 0.21x

Du Pont Decomposition

ROE = Leverage (4.502) x Asset Turnover (0.741) x Op. Margin (20.05%) x Debt Burden (0.896) = 59.94%
ROA = Asset Turnover (0.741) x Op. Margin (20.05%) = 14.85% [Direct: ROE 59.71%, ROA 14.9% – verified]

C. Key Findings

Performance and Profitability

CAT is creating substantial value. EVA of \$9.5B confirms that after-tax operating income (\$13.0B) far exceeds the capital charge (\$3.5B at 7.9% WACC), and MVA of \$152.5B reflects the market's confidence in continued above-cost returns. ROC of 29.5% and a 20% operating margin are well above industry norms, driven by CAT's aftermarket parts mix and pricing power. ROE of 59.7% is high but meaningfully amplified by leverage — it should be read alongside the Du Pont breakdown.

Finding 1 — Strong value creation across all performance metrics. ROC of 29.5% is roughly double the cost of capital.

Efficiency

Days in Inventory of 140 days is the clearest operational gap — high for capital goods manufacturers where 80-110 days is typical. Reducing to 110 days would free roughly \$3.5B in working capital (30 days x \$118.5M daily COGS). Asset turnover of 0.74x and a 52-day collection period are in line with the business model. Free cash flow was \$12.5B (\$14.2B operating minus \$1.7B capex), providing strong internal funding capacity.

Finding 2 — Inventory at 140 days is the primary operational opportunity. Cutting to 110 days frees ~\$3.5B.

Leverage and Liquidity

Total debt at 77.8% of assets looks elevated, but much of this sits in Cat Financial, whose debt is matched by finance receivables. TIE of 8.28x and cash coverage of 9.11x confirm comfortable interest coverage. LT debt rose \$2.9B in FY2024, driven by \$7.8B in buybacks. The quick ratio of 0.50x is the key liquidity concern — inventory cannot be liquidated quickly in a downturn, and CAT's revenue is cyclically tied to mining and construction capex.

Finding 3 — Coverage ratios are healthy, but the 0.50x quick ratio and rising debt from buybacks warrant monitoring.

D. Du Pont Analysis

Margin is the main driver. A 20% operating profit margin — exceptional for a cyclical manufacturer — contributes more to ROE than any other factor. Leverage (4.50x) amplifies the underlying 14.9% ROA into 59.7% ROE; without it, returns would still be strong but far less pronounced. Asset turnover (0.74x) is constrained by the capital-intensive model but can improve through inventory reduction, which would lift ROE without adding financial risk. The debt burden of 0.896 shows that financing costs absorb about 10% of operating income before reaching net income — a reasonable cost given the leverage benefit.

E. Strategic Recommendations

1. Cut Days in Inventory from 140 to 110 days

This would release ~\$3.5B in cash through operational improvement alone. Review demand-driven production scheduling, SKU rationalization, and dealer stocking policies.

2. Publish EVA and MVA in investor materials

With \$7.8B in FY2024 buybacks under scrutiny, a formal EVA bridge at investor day would show that capital allocation is generating returns well above the cost of capital.

3. Add a WACC sensitivity table to EVA reporting

Every 100bp shift in WACC moves EVA by ~\$440M. Show the board EVA across 6%-10% to stress-test the value-creation conclusion and prepare for rate volatility.

4. Set a minimum cash target of \$8-9B

The 0.50x quick ratio and \$2.9B increase in LT debt leave less liquidity cushion than in prior years. A formal floor and pre-approved commercial paper facility would reduce exposure to a cyclical revenue decline.

F. Structured AI Prompt

AI Prompt: Write a completed professional executive memo that references a complete ratio spreadsheet containing all the data listed below with each section having its own tab in Excel, summarize that data, and provide strategic recommendations based on the information provided. Follow named range principles listed below and do not infer values. Ask additional questions before proceeding.

GOAL

Create a Microsoft Excel workbook with 30+ accounting ratios for Caterpillar Inc. (NYSE: CAT), FY2024. Use named ranges exclusively – no bare cell references in the Ratios tab. Five sheets: Balance Sheet, Income Statement, Cash Flow, Ratios, Notes. Colors: Yellow=Inputs | Blue=Assumptions | Green=Formulas | Gray=Outputs

COMPANY FINANCIAL DATA (all values in \$M USD)

BALANCE SHEET

Named Range	FY2024	FY2023
BAL_cash_marketable_securities_2024	6,889	–
BAL_receivables_2024	9,282	
BAL_receivables_2023		9,310
BAL_inventories_2024	16,827	
BAL_inventories_2023		16,565
BAL_assets_current_2024	45,682	–
BAL_assets_total_2024	87,764	
BAL_assets_total_2023		87,476
BAL_liabilities_current_2024	32,272	–
BAL_liabilities_total_2024	68,270	–
BAL_debt_long_term_2024	27,351	
BAL_debt_long_term_2023		24,472
BAL_equity_shareholders_2024	19,494	
BAL_equity_shareholders_2023		19,503

INCOME STATEMENT

INC_sales=64,809 INC_cost_goods_sold=43,244 INC_ebit=14,187
INC_interest_expense=1,713 INC_depreciation=1,413 INC_net=11,646

CASH FLOW

CASH_operating=14,184 CASH_capex=(1,702)

MARKET & ANALYST INPUTS

share_price=349.58 shares_outstanding=492M cost_capital=0.079 tax_rate=0.214

DERIVED INTERMEDIATES (Green, computed in Ratios tab)

market_cap = share_price x shares_outstanding
startYear_equity = BAL_equity_shareholders_2023
startYear_inventory = BAL_inventories_2023
startYear_receivables = BAL_receivables_2023
startYear_total_assets = BAL_assets_total_2023
startYear_total_cap = BAL_debt_long_term_2023 + BAL_equity_shareholders_2023
currentYear_after_tax_op_inc = INC_net + (1 - tax_rate) x INC_interest_expense
currentYear_daily_sales = INC_sales / 365
currentYear_daily_cogs = INC_cost_goods_sold / 365
currentYear_total_cap = BAL_debt_long_term_2024 + BAL_equity_shareholders_2024
currentYear_nwc = BAL_assets_current_2024 - BAL_liabilities_current_2024
avg_equity = AVERAGE(startYear_equity, BAL_equity_shareholders_2024)
avg_total_assets = AVERAGE(startYear_total_assets, BAL_assets_total_2024)
avg_total_cap = AVERAGE(startYear_total_cap, currentYear_total_cap)

RATIO FORMULAS

PERFORMANCE

MVA = market_cap - BAL_equity_shareholders_2024
M/B = market_cap / BAL_equity_shareholders_2024
EVA = currentYear_after_tax_op_inc - (cost_capital x startYear_total_cap)

PROFITABILITY (compute start-of-year and average-based variants)

ROA = currentYear_after_tax_op_inc / startYear_total_assets
ROC = currentYear_after_tax_op_inc / startYear_total_cap
ROE = INC_net / startYear_equity

[AVG variants use avg_ prefix denominators]

EFFICIENCY (* = assign named range for Du Pont)

RATIO_asset_turnover* = INC_sales / startYear_total_assets
Receivables turnover = INC_sales / startYear_receivables
Avg collection period = startYear_receivables / currentYear_daily_sales
Inventory turnover = INC_cost_goods_sold / startYear_inventory
Days in inventory = startYear_inventory / currentYear_daily_cogs
Profit margin = INC_net / INC_sales
RATIO_operating_profit_margin* = currentYear_after_tax_op_inc / INC_sales

LEVERAGE (* = assign named range for Du Pont)

LT debt ratio = BAL_debt_long_term_2024 / currentYear_total_cap
LT debt-equity ratio = BAL_debt_long_term_2024 / BAL_equity_shareholders_2024
Total debt ratio = BAL_liabilities_total_2024 / BAL_assets_total_2024
Times interest earned = INC_ebit / INC_interest_expense
Cash coverage = (INC_ebit + INC_depreciation) / INC_interest_expense
RATIO_debt_burden* = INC_net / currentYear_after_tax_op_inc
RATIO_leverage* = BAL_assets_total_2024 / BAL_equity_shareholders_2024

LIQUIDITY

NWC to assets = currentYear_nwc / BAL_assets_total_2024
Current ratio = BAL_assets_current_2024 / BAL_liabilities_current_2024
Quick ratio = (BAL_cash Marketable_securities_2024 + BAL_receivables_2024) / BAL_liabilities_current_2024
Cash ratio = BAL_cash Marketable_securities_2024 / BAL_liabilities_current_2024

DU PONT

ROA (Du Pont) = RATIO_asset_turnover x RATIO_operating_profit_margin
ROE (Du Pont) = RATIO_leverage x RATIO_asset_turnover
x RATIO_operating_profit_margin x RATIO_debt_burden

VERIFICATION CHECKS

1. BS FY2024: \$87,764M = \$68,270M + \$19,494M (diff = \$0)
2. BS FY2023: \$87,476M = \$67,973M + \$19,503M (diff = \$0)
3. Du Pont ROA matches direct ROA within 0.01 – expected ~14.85%
4. Du Pont ROE matches direct ROE within 1% – expected ~59.7-59.9%
5. EVA: \$12,992M - (7.9% x \$43,975M) = \$9,518M

NOTES TAB – document these four assumptions

- (a) WACC 7.9%: no CAPM build-up; add sensitivity table 6%-10%
 - (b) Total cap excludes short-term debt (\$4.4B) + current LTD (\$6.7B)
 - (c) Quick ratio excludes Cat Financial receivables (\$9.6B)
 - (d) Tax rate 21.4% statutory; effective rate ~20.2% – derive from financials in future
- Include: data source, color-coding legend, formula correction log.

G. Executive Justification

Financial Health and Value Creation

CAT generated \$12.5B in free cash flow in FY2024 — enough to fund capex, pay \$2.7B in dividends, and repurchase \$7.8B in shares while holding \$6.9B in cash. EVA of \$9.5B and ROC of 29.5% confirm that capital is being put to work at returns roughly double the cost of capital. A 20% operating margin in a cyclical industry reflects genuine structural advantages: the global dealer network, Cat Financial's captive financing, and aftermarket parts pricing power.

Operational and Capital Structure Priorities

Inventory is the clearest near-term lever — 140 days is high for the sector, and reducing it would free \$3.5B without any financial engineering. On the capital structure side, headline leverage ratios overstate risk because Cat Financial inflates consolidated debt; TIE of 8.28x and cash coverage of 9.11x show debt is well-serviced. Still, the \$2.9B rise in LT debt from buybacks narrows the balance sheet buffer and should be communicated clearly to rating agencies.

Liquidity and Reporting

The 0.50x quick ratio is the most pressing balance sheet vulnerability. CAT's cyclical nature means liquidity pressure and earnings pressure arrive together; a formal cash floor of \$8-9B and a pre-approved credit facility would reduce that risk. On reporting, four model assumptions deserve disclosure: WACC sensitivity (each 100bp shifts EVA by ~\$440M),

capitalization scope (excludes \$11.1B of short-term debt), quick ratio composition (Cat Financial excluded), and the gap between the 21.4% statutory tax rate and the 20.2% effective rate implied by the income statement.

Data source: Caterpillar Inc. 10-K FY2024, SEC EDGAR. All figures in millions of USD unless noted. Van Brussel, BUS-314, Shidler College of Business, April 2026.